

FROM SINGLE SYNAPSES TO LARGE-SCALE SPIKING NEURAL NETWORK SIMULATIONS WITH NEST AND NESTML

Charl Linssen & Sandra Diaz | March 13th, 2026 | EBRAINS Student Conference

Human Brain Project
EBRAINS
Co-funded by the European Union
JÜLICH Forschungszentrum
Member of the Helmholtz Association

Welcome!

09:00	Overview and introduction to NEST Simulator *Charl Linssen*
09:15	Interactive network creation and simulation with NEST Desktop
09:30	Simulation and data analysis via Jupyter Notebooks
10:00	Network homeostasis through structural synaptic plasticity *Sandra Diaz*
10:30	Creating custom neuron and synapse models using NESTML: introduction *Charl Linssen*
10:45	Custom spike-frequency adaptation models in NESTML
11:30	Closing

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Prerequisites: Python programming and JupyterLab

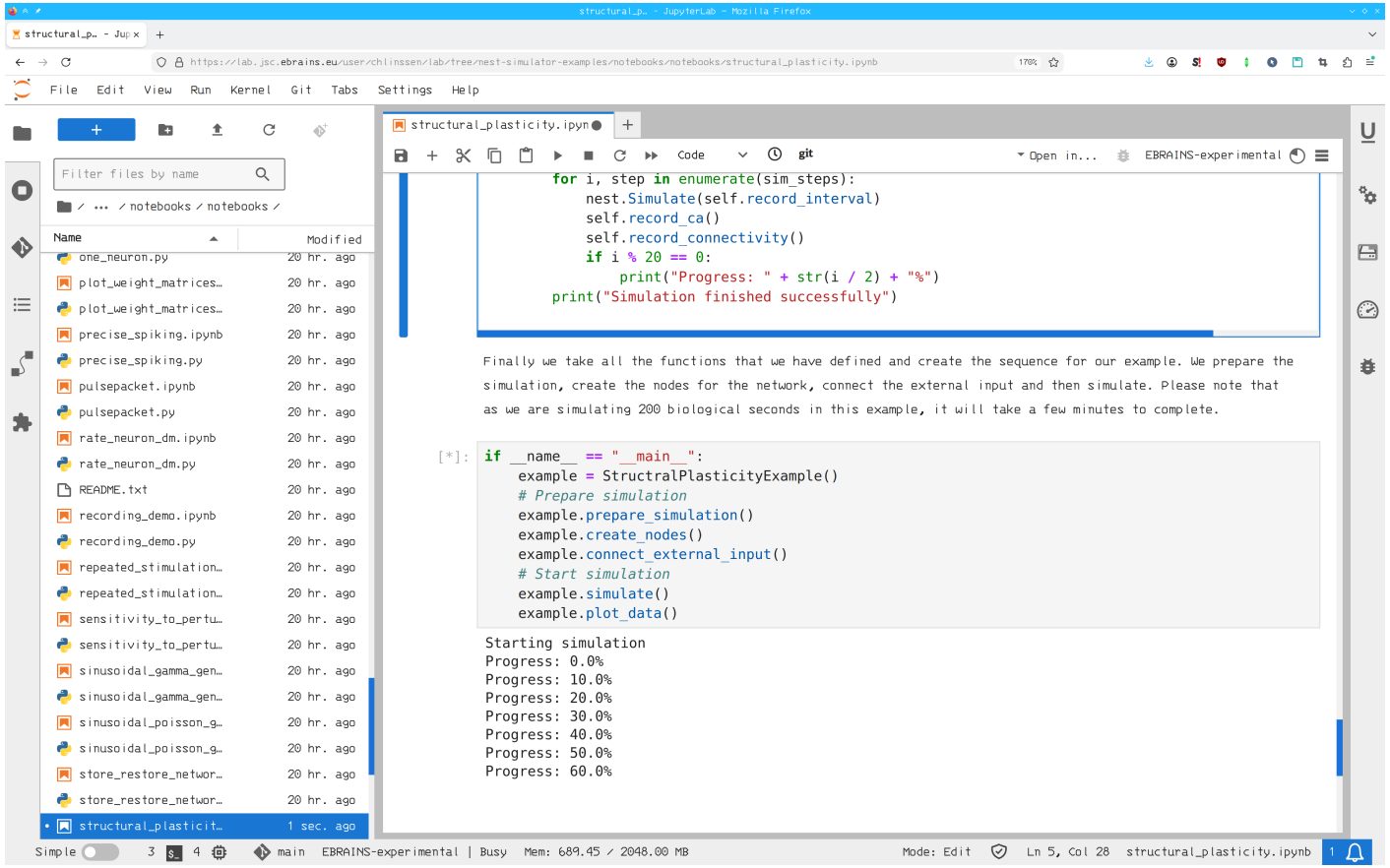
JupyterLab is an interactive graphical user interface (GUI) for the Python programming language.

Usage tips:

- Shift+Return runs a single cell.
- Use the ⏩ icon in the toolbar to run all cells.

For an advanced JupyerLab tutorial, see e.g. https://realpython.com/using-jupyterlab/

Code will run inside an EBRAINS VM with 2 GB RAM. Files stored in your VM home directory can disappear—save them locally for safekeeping!



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Interactive network creation and simulation with NEST Desktop

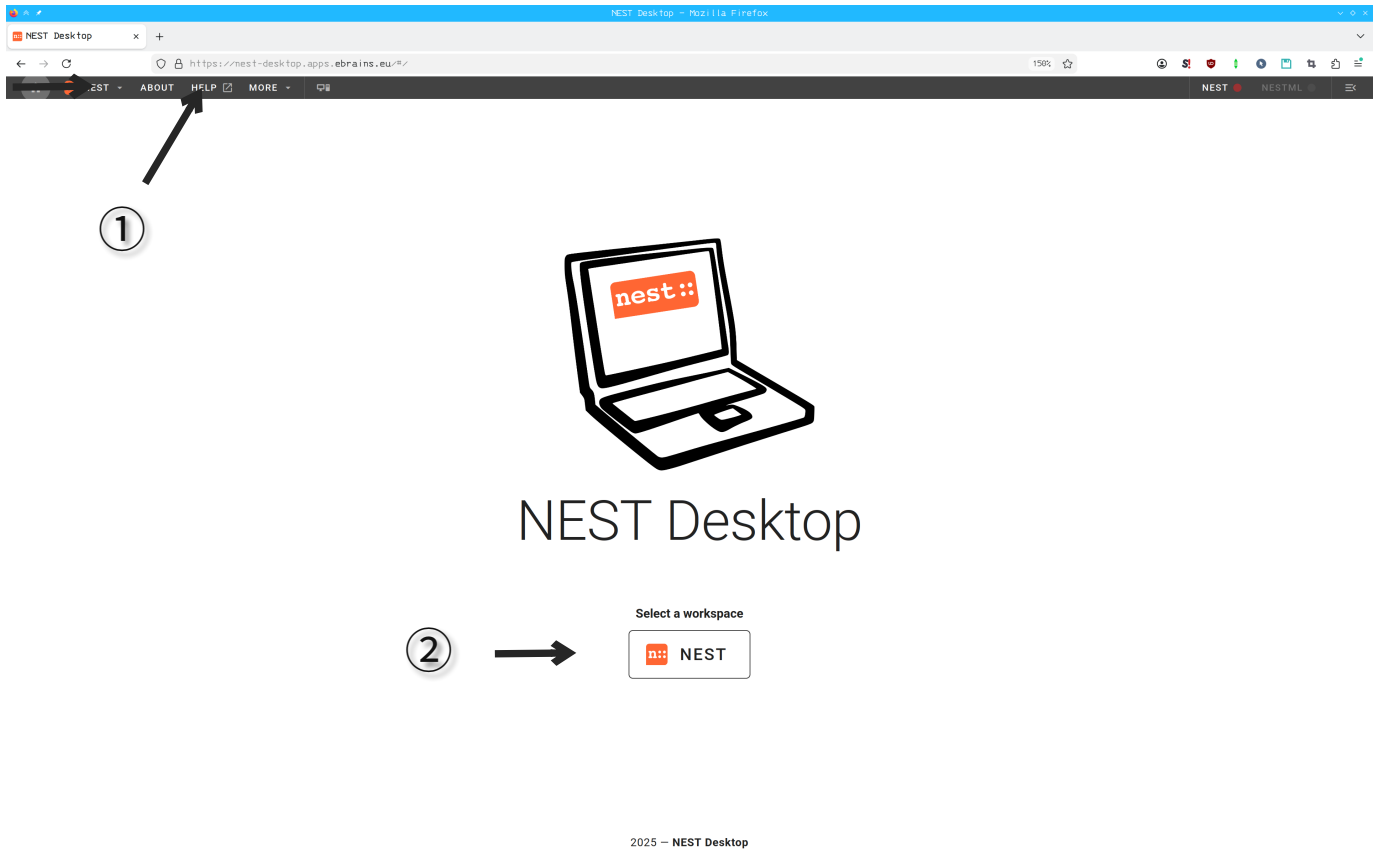
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Running NEST Desktop

① Use your browser to go to the URL:

https://nest-desktop.apps.ebrains.eu/

② Select NEST workspace



NEST Desktop stores models as cookies in your browser. Models will disappear when your browser cleans up cookies. Always save your models locally for safekeeping!

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Tutorial materials for NEST Desktop

- Central place for NEST Desktop documentation:

  https://nest-desktop.readthedocs.io/

- Background reading about balanced networks (see especially figures 5 and 8):

  https://github.com/clinssen/NEST-workshop/blob/main/materials/nest-desktop/brunel2000.pdf

- For the step-by-step tutorial about creating a neuron and a balanced network:

  https://github.com/clinssen/NEST-workshop/.../nest-desktop/NEST_Desktop_tutorial_steps.docx

Running the balanced network tutorial notebook

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Running the balanced network tutorial notebook

① Use your browser to go to the URL:

https://lab.ebrains.eu/

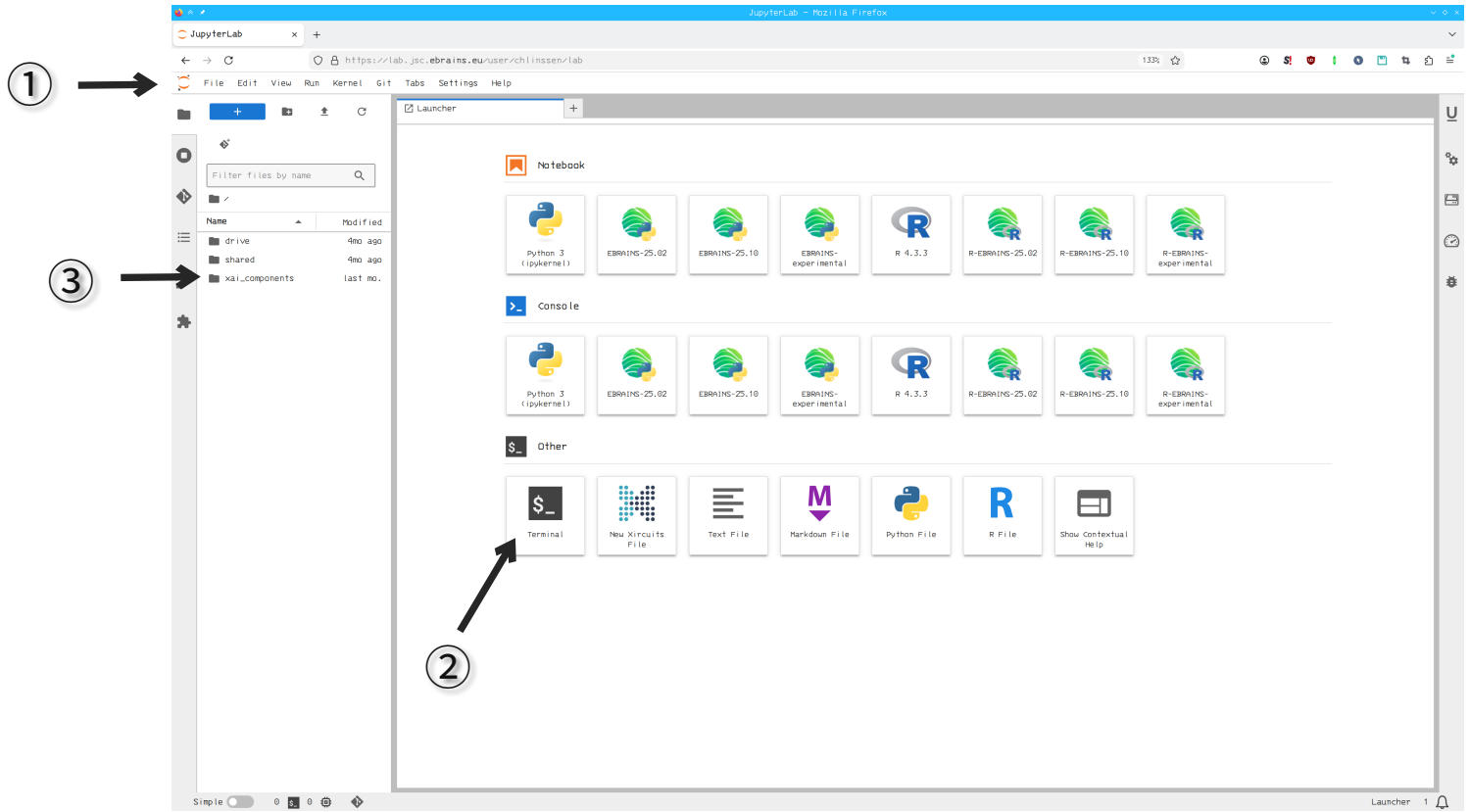
Select JSC as execution site.

② Open a terminal and run:

```
wget
https://github.com/clinssen/NEST-
workshop/raw/refs/heads/main/mate
rials/nest/
nest_intro_and_brunel/NEST
%20balanced%20network
%20tutorial.ipynb
```

③ Open the downloaded notebook

④ Select the “EBRAINS Experimental” kernel



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Running the balanced network tutorial notebook

① Use your browser to go to the URL:

https://lab.ebrains.eu/

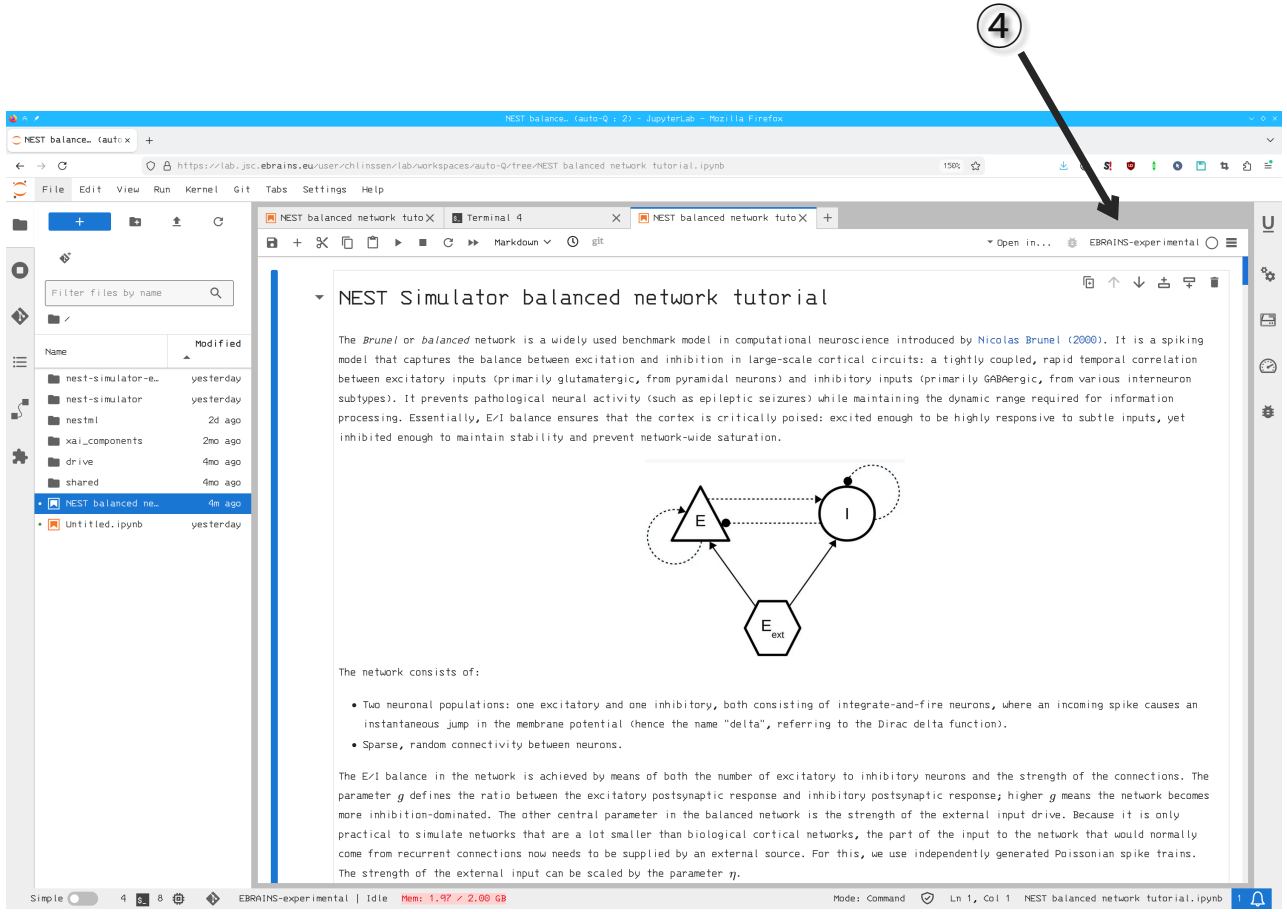
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Network homeostasis through structural synaptic plasticity

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Running the NEST structural plasticity tutorial

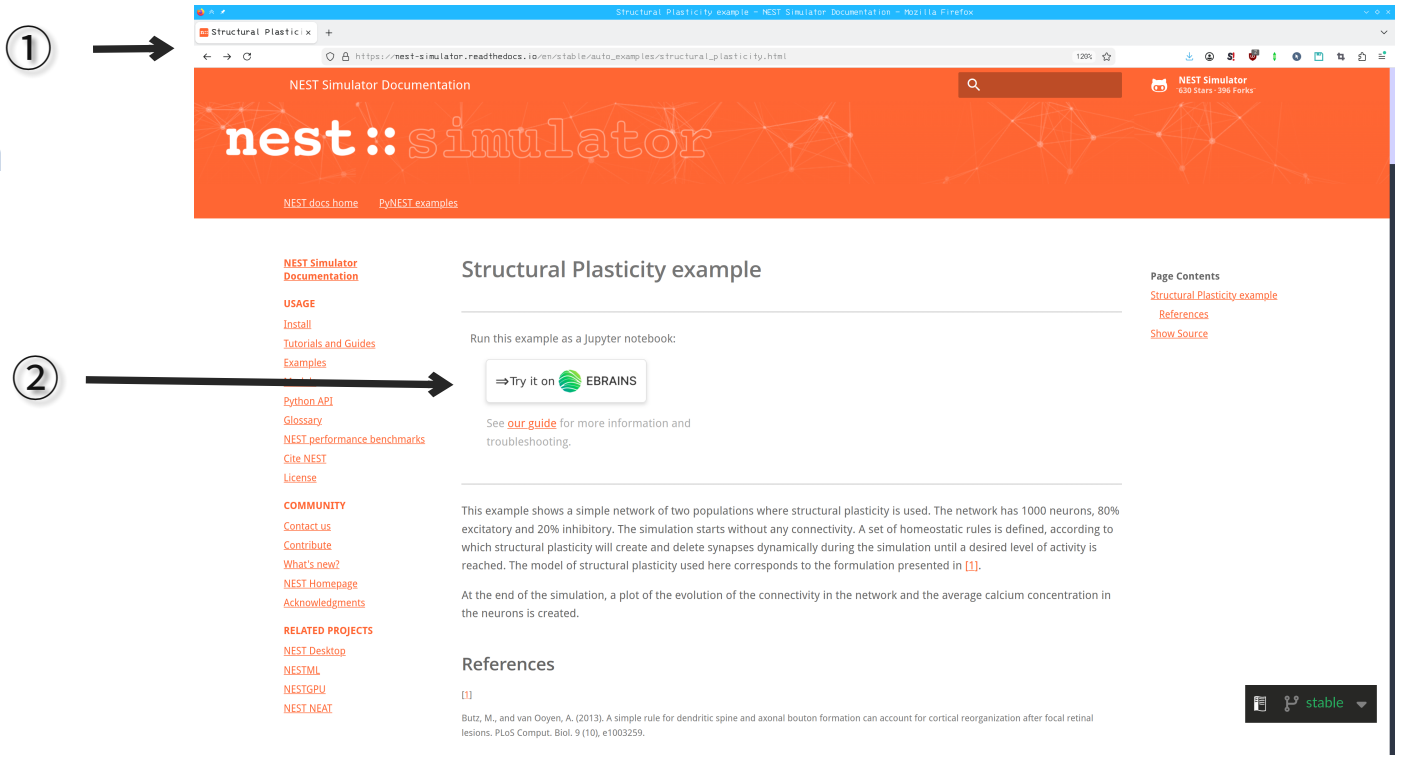
① Use your browser to go to the URL:

https://nest-simulator.readthedocs.io/en/stable/auto_examples/structural_plasticity.html

② Click the “Try it on EBRAINS” button.

③ Select JSC as execution site.

④ Select the “EBRAINS Experimental” kernel



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Running the NEST structural plasticity tutorial

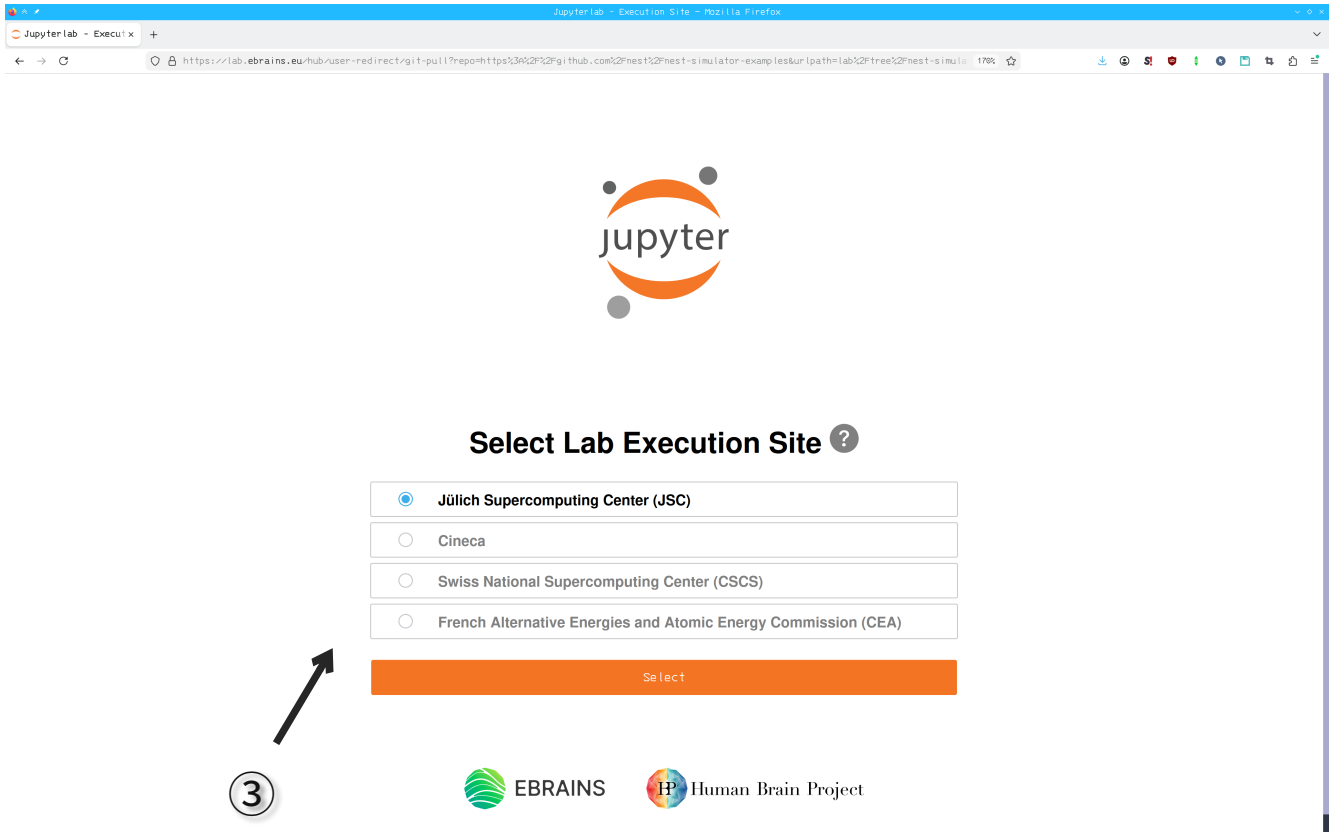
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Running the NEST structural plasticity tutorial

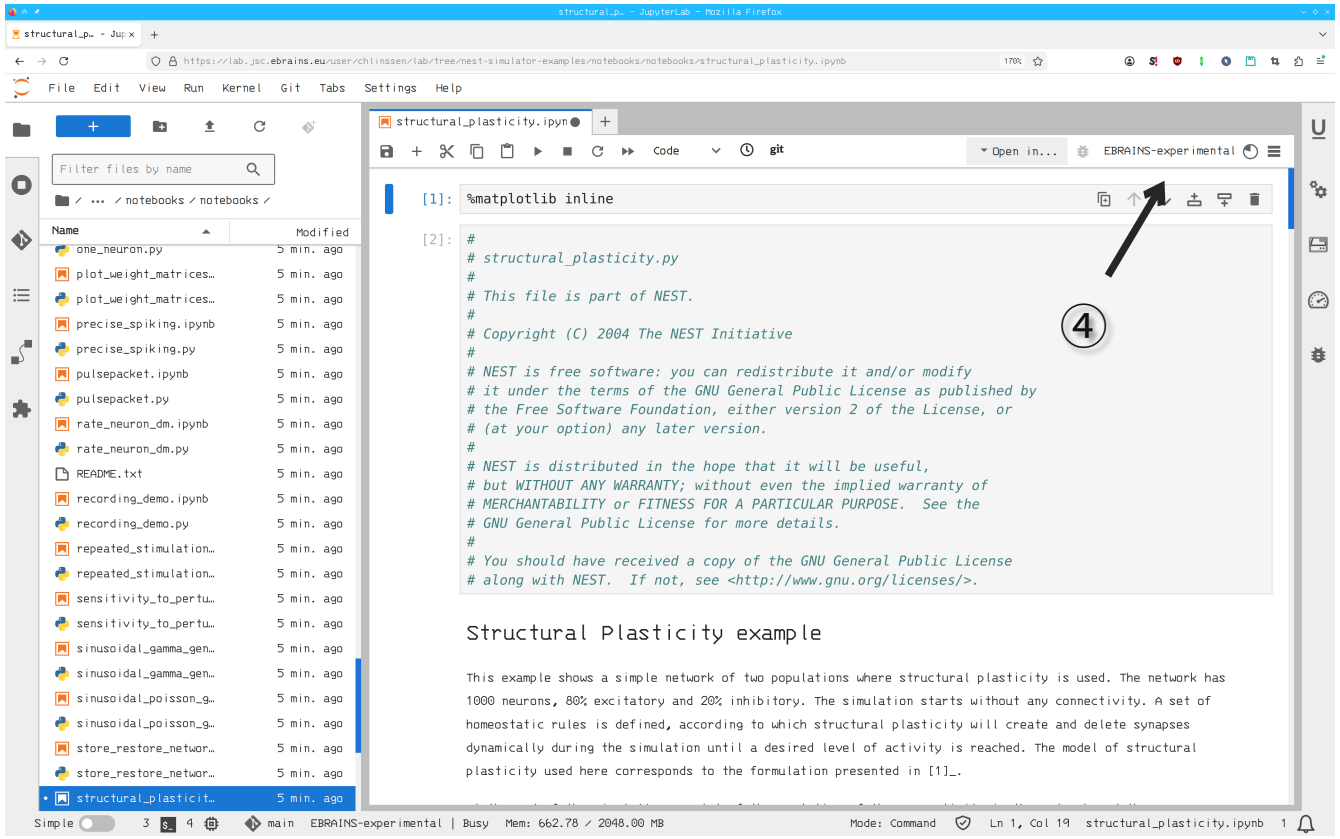
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Tutorial materials for NEST Simulator

- Central place for NEST Simulator documentation:

  https://nest-simulator.readthedocs.io/

  See especially the “examples” section.

Creating custom neuron and synapse models using NESTML

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Running NESTML

① Use your browser to go to the URL:

https://lab.ebrains.eu/

Select JSC as execution site.

② Open a terminal and run:

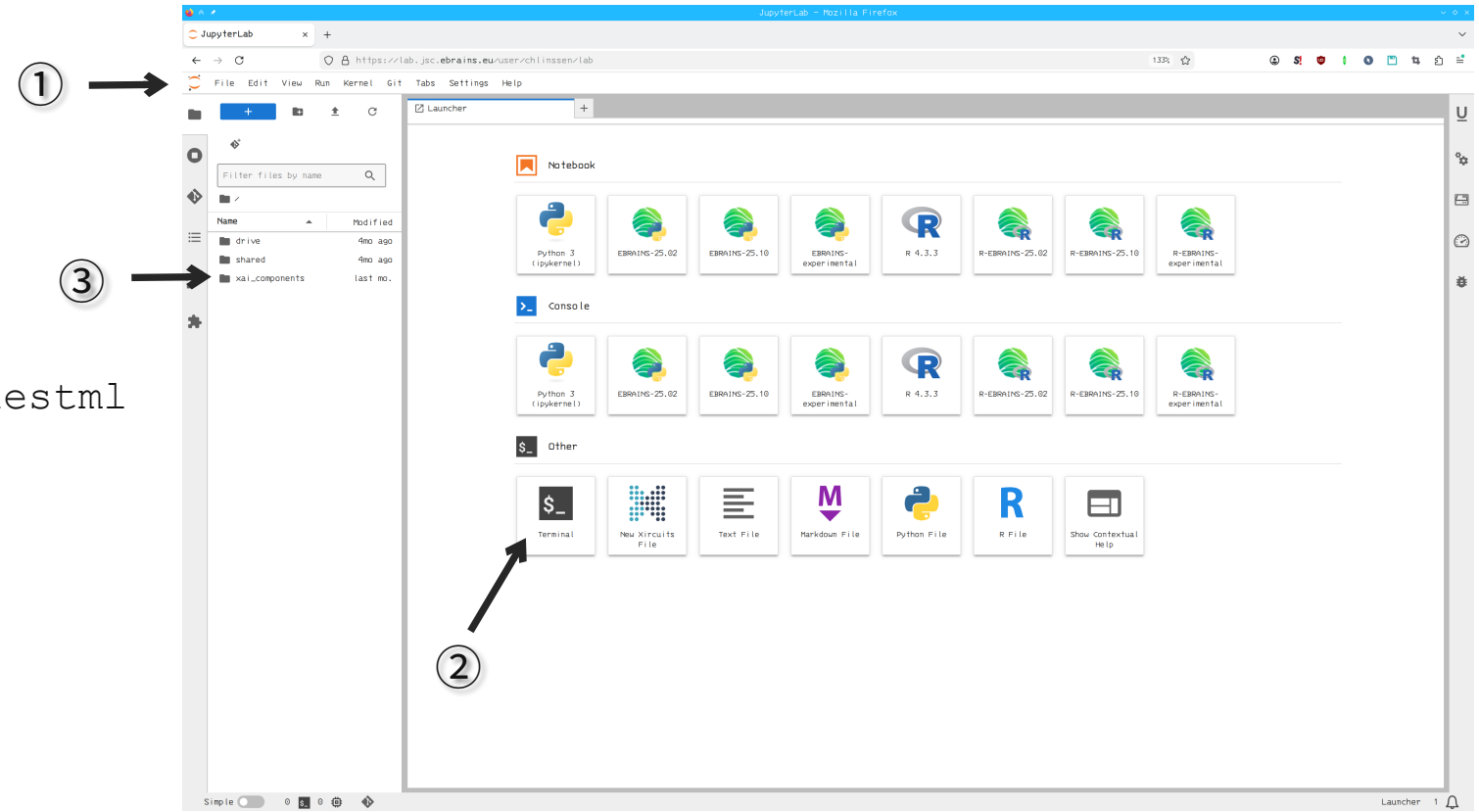
```
git clone https://github.com/nest/nestml
cd nestml
git checkout v8.2.0
```

③ Go back to the start page and open a tutorial notebook:

```
nestml/doc/tutorials/...
```

We will start with the “Spike frequency adaptation” notebook.

④ Select the “EBRAINS Experimental” kernel



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Running NESTML

① Use your browser to go to the URL:

https://lab.ebrains.eu/

Select JSC as execution site.

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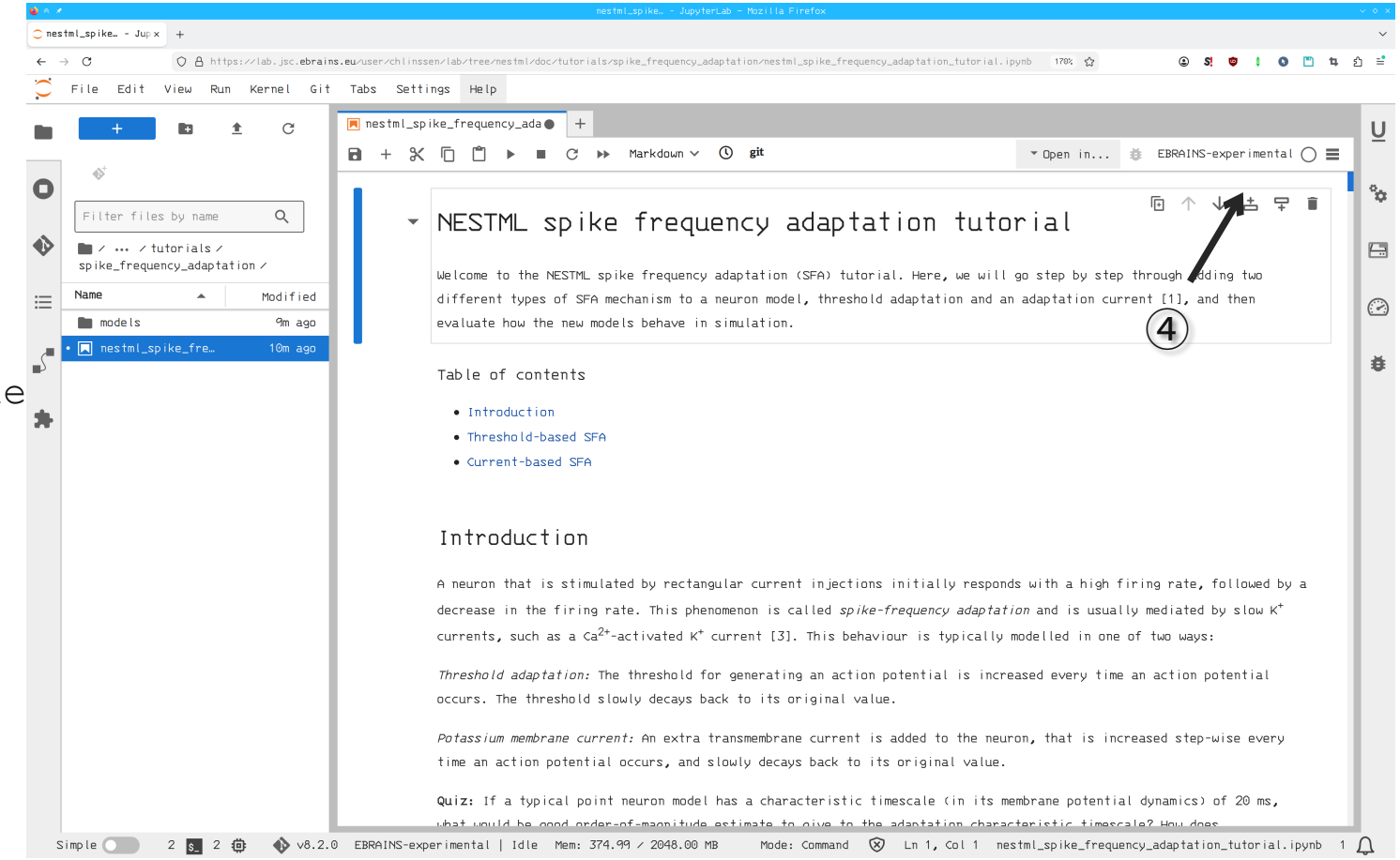
```
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cd nestml
git checkout v8.2.0
```

③ Go back to the start page and open a tutorial notebook:

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Tutorial materials for NESTML

- Central place for NESTML documentation:

  https://nestml.readthedocs.io/

- All available tutorials are in the `nestml/doc/tutorials/` directory that you git cloned.

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Thank you for your participation!

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https://www.surveymonkey.com/r/L5L72YB

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